W6.2 Interaction

Lecture

Agenda

- Filtering
- Animations
- Tooltips

Interactivity

Allows exploration:

- Large data sets
- Data sets with multiple attributes

Choose your own adventure rather than being locked into author's story

Basic interactions

- Filtering
- Tool tips
- Animations

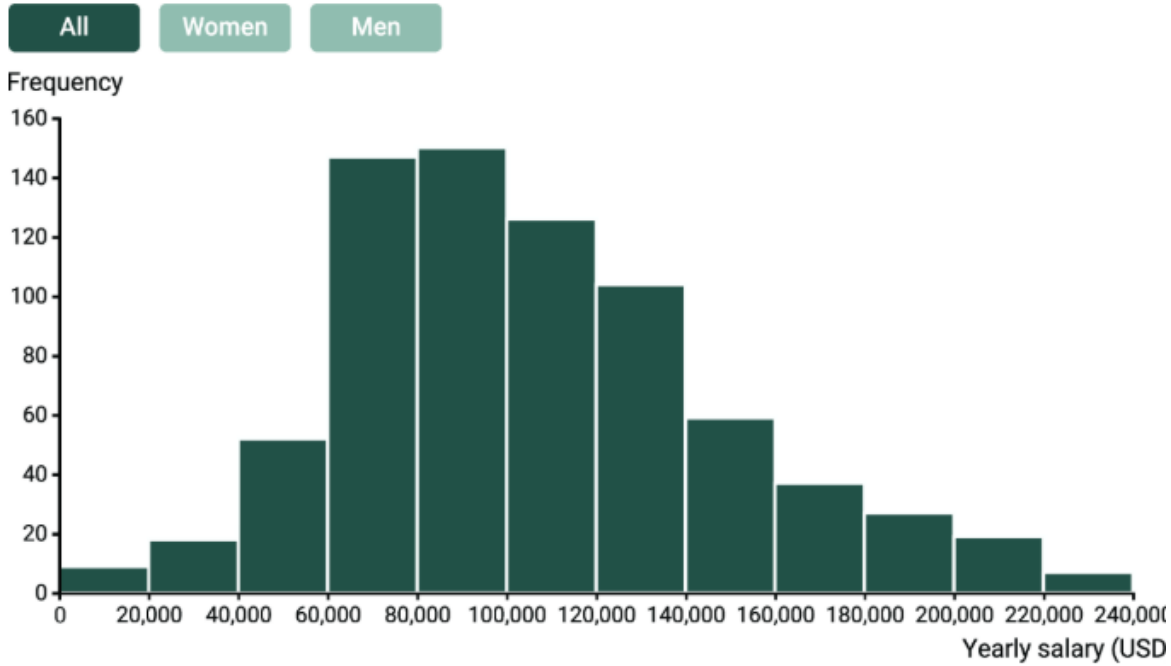


Figure 7.1 Histogram of the salary distribution among data visualization practitioners in the US with buttons to filter the chart by gender

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Setup

Add new file for interaction code

- interactions.js

index.html

- Add call to interactions.js
- Add div for filters (after chart call)

```
<div id="filters"></div>
```

load-data.js

- Add call to filters function so it has access to data

```
// Load data
d3.csv("./data/screen_time.csv", d => {
  return {
    Age: +d.Age, // Convert Age to a number
    Gender: d.Gender,
    ScreenTimeType: d.Screen_Time_Type,
    DayType: d.Day_Type,
    AverageScreenTime: +d.Average_Screen_Time, // Convert to a number
    SampleSize: +d.Sample_Size // Convert to a number
  };
}).then(data => {
  console.log("Processed Data:", data);

  drawHistogram(data);
  populateFilters(data);

});
```


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Setup

Add constants needed for interaction to shared-constants

Update histogram to use scale shared globally in shared-constants

xScale
.domain([minTime, maxTime])
.range([0, innerWidth]);

yScale
.domain([0, binsMaxLength])
.range([innerHeight, 0])
.nice();

shared-constants.js > ...

```
/*******************************/
/*    Declare the constants    */
/*******************************/
const margin = {top: 40, right: 30, bottom: 50, left: 40};
const width = 1000;
const height = 500;
const innerWidth = width - margin.left - margin.right;
const innerHeight = height - margin.top - margin.bottom;


/*****************************************/
/*  Make the bin Generator accessible globally  */
/*****************************************/
const binGenerator = d3.bin() // Create a bin generator
  .value(d => d.average_screen_time);

/*****************************************/
/*  Make the scales accessible globally  */
/*****************************************/
const xScale = d3.scaleLinear();
const yScale = d3.scaleLinear();

/*****************************************/
/*  Make the colours accessible globally  */
/*****************************************/
const slateGray = "#305252";
const gray = "#606464";
const white = "#faffff";

/*****************************************/
/*  Make the filter options accessible globally  */
/*****************************************/
const filters_gender = [
  { id: "all", label: "All", isActive: true },
  { id: "female", label: "Female", isActive: false },
  { id: "male", label: "Male", isActive: false },
  { id: "other_prefer_not_to_say", label: "Other/Prefer Not to Say", isActive: false }
];
```


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Filtering

Define filtering options with array

(in **shared-constants.js**)

```
/*****************************************/
/*  Make the filter options accessible globally  */
/*****************************************/
const filters_gender = [
    { id: "all", label: "All", isActive: true },
    { id: "female", label: "Female", isActive: false },
    { id: "male", label: "Male", isActive: false },
    { id: "other_prefer_not_to_say", label: "Other/Prefer Not to Say", isActive: false }
];
```

```
const filters = [
  { id: "all", label: "All", isActive: true },
  { id: "female", label: ”Female", isActive: false },
  { id: "male", label: ”Male", isActive: false },
];
```

id – identifier for each filter option

label: human readable, displayed in user interface

isActive: boolean value indicating whether filter is active or not

- Default state – ‘all” active, “female” and “male” inactive)

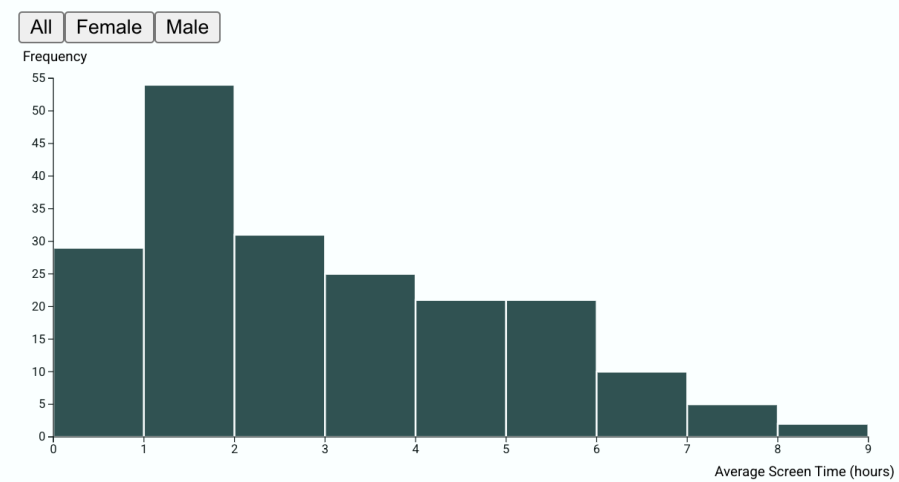
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Add filter buttons (in interactions.js)

```
const populateFilters = (data) => {

    d3.select("#filters")
      .selectAll(".filter")
      .data(filters)
      .join("button")
        .attr("class", d => `filter ${d.isActive ? "active" : ""}`)
        .text(d => d.label);

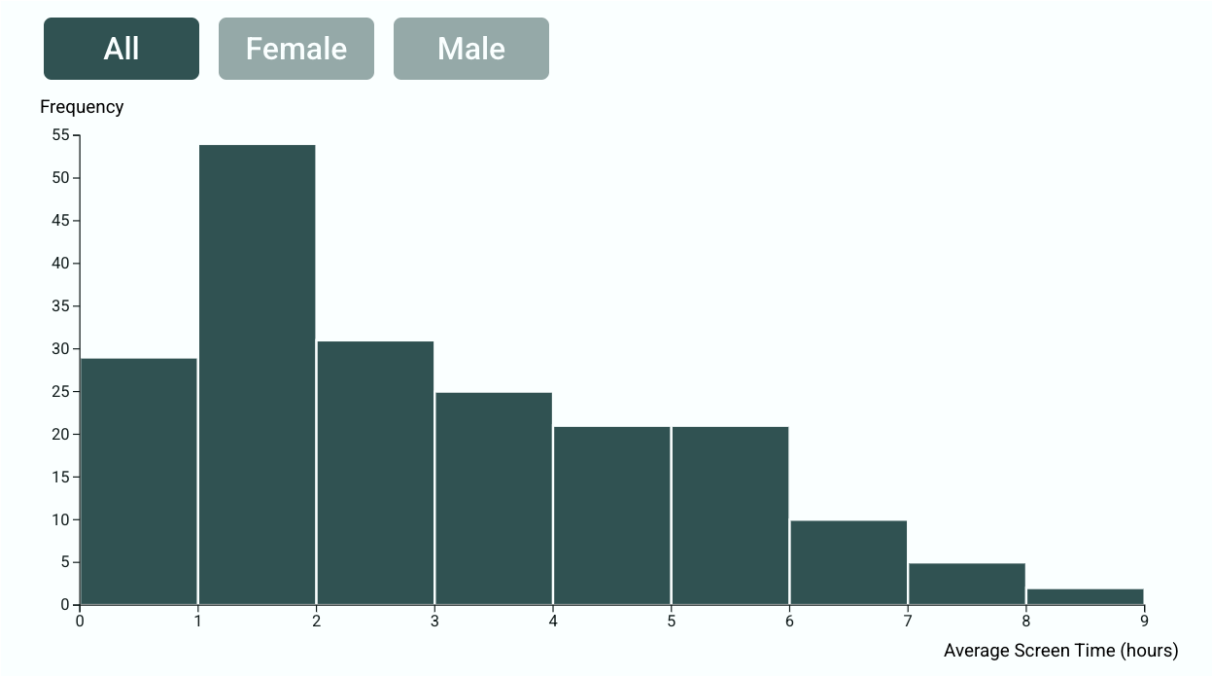
};
```



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Find some nice css to make buttons look good...

```css
/* Filters */
#filters {
  margin: 10px 0px 5px;
}
.filter {
  min-width: 80px;
  margin: 0 5px;
  padding: 8px 10px;
  font-family: 'Roboto', sans-serif;
  font-size: 16px;
  line-height: 1;
  color: #faffff;
  font-weight: 500;
  background-color: rgba(48, 82, 82, 0.5);
  border: none;
  border-radius: 4px;
  cursor: pointer;
  transition: background-color 0.3ms ease-in;
}
.filter:hover,
.filter.active {
  background-color: rgba(48, 82, 82, 1);
  transition-timing-function: ease-out;
}
.filter.active {
  cursor: default;
}
```



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Capture user events

To make interactive, need to add JavaScript event listener

Event listener just sits and waits for specified event to happen (mouse over, click etc)!

```
selection.on("event type",     (e, d) => { ... });
```

(string)

"click"
"mouseover"
"touch"
"keydown"
...

(callback function)

Parameters:
  e: Detected DOM event.
  d: Datum attached to the selection.

Perform the desired action(s between the brackets.

Figure 7.2 The `on()` method attaches an event listener to a D3 selection and performs specific actions when the event occurs.

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Add buttons and event listener

```
d3.select("#filters")
  .selectAll(".filter")
  .data(filters)
  .join("button")
    .attr("class", d => `filter ${d.isActive ? "active" : ""}`)
    .text(d => d.label)

    .on("click", (e, d) => {
      console.log("DOM event", e);
      console.log("Attached datum", d);
    });
```

```
/****************************************/
/*  Make the filter options accessible globally  */
/****************************************/
const filters = [
    { id: "all", label: "All", isActive: true },
    { id: "female", label: "Female", isActive: false },
    { id: "male", label: "Male", isActive: false },
  ];
```

html created by D3 to make buttons

```
<div id="filters_gender">
  <button class="filter active">All</button>
  <button class="filter">Male</button>
  <button class="filter">Female</button>
</div>
```

CSS

```
/* Filters */
#filters {
  margin: 10px 0px 5px;
}
.filter {
  min-width: 80px;
  margin: 0 5px;
  padding: 8px 10px;
  font-family: 'Roboto', sans-serif;
  font-size: 16px;
  line-height: 1;
  color: #faffff;
  font-weight: 500;
  background-color: rgba(48, 82, 82, 0.5);
  border: none;
  border-radius: 4px;
  cursor: pointer;
  transition: background-color 0.3ms ease-in;
}
.filter:hover,
.filter.active {
  background-color: rgba(48, 82, 82, 1);
  transition-timing-function: ease-out;
}
.filter.active {
  cursor: default;
}
```

DOM – object has lots of info about. event

```
DOM event ▸ click { target: button.filter , buttons: 0, clientX: 169, clientY: 39, layerX: 169, layerY: 245 }

Attached datum ▾ Object { id: "female", label: "Female", isActive: false }
                   id: "female"
                   isActive: false
                   label: "Female"
                 ▸ <prototype>: Object { … }
```


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Add filter conditionals

```
.on("click", (e, d) => {

    if (!d.isActive) {
        // Update the isActive state for all filters
        filters.forEach(filter => {
            filter.isActive = d.id === filter.id; // Set true only for the clicked filter
        });
    // Update the filter buttons' active class
    d3.selectAll("#filters .filter")
        .classed("active", filter => filter.isActive);

    updateHistogram(d.id, data);
}
```

Click female button updates isActive to true

```
Attached datum ▶ Object { id: "female", label: "Female", isActive: true }
```

"female" is passed to updateHistogram via d.id

```
{ id: "female", label: "Female", isActive: true }
```


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D3 .classed() method

Used to dynamically add, remove, or toggle CSS classes on selected elements based on a condition.

- Add a CSS class to elements if a condition is true.
- Remove a CSS class from elements if a condition is false.

In our code it adds or removes class “active”

```
.classed("active", filter => filter.isActive);
```

```
selection.classed("class name",      value);
```

(string)	(boolean)
"my-class-name"	true false

Figure 7.3 The `classed()` method provides control over each class name of a selection separately. Its first argument is the class name, and its second is true or false, indicating whether the selection should have the class name or not.

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Create new function to update histogram

```
const updatedData = (() => {
    if (filterId === "all") {
        return data;
    } else {
        return data.filter(respondent => respondent.gender === filterId);
    }
})();
```

```
const updateHistogram = (filterId, data) => {

    const updatedData = filterId === "all"
        ? data
        : data.filter(respondent => respondent.gender === filterId);

    const updatedBins = binGenerator(updatedData);

      d3.selectAll("#histogram rect")
        .data(updatedBins)
        .attr("y", d => yScale(d.length))
        .attr("height", d => innerHeight - yScale(d.length));
};
```

Remember – binGenerator and x and y Scales need to be in shared constants

```
/*****************************************/
/*  Make the bin Generator accessible globally  */
/*****************************************/
  const binGenerator = d3.bin() // Create a bin genera
    .value(d => d.average_screen_time);

/*****************************************/
/*  Make the scales accessible globally  */
/*****************************************/
const xScale = d3.scaleLinear();
const yScale = d3.scaleLinear();
```


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Animations

Transition types

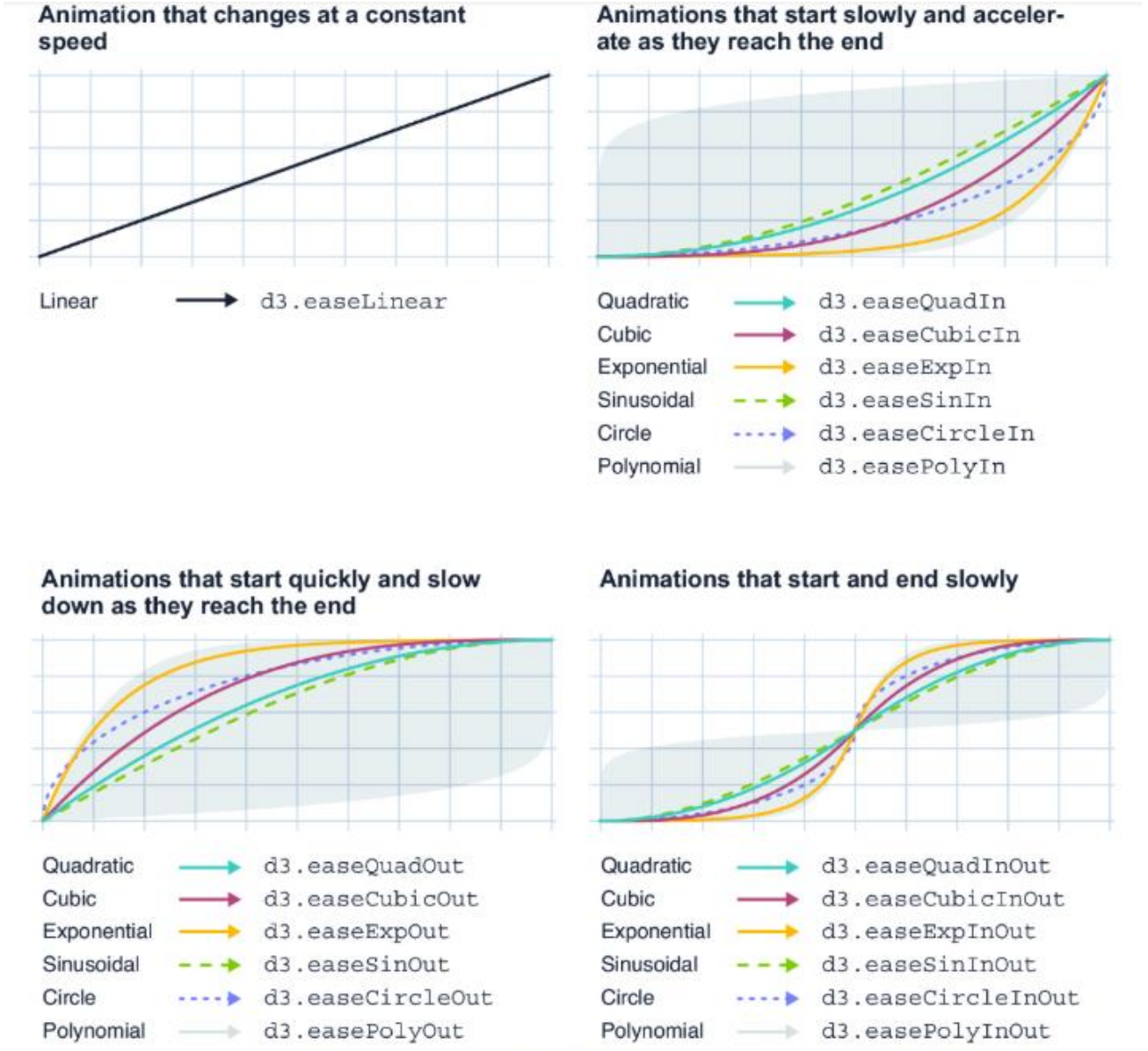


Figure 7.5 The most common easing functions in comparison to a linear rate of change. On each graph, the horizontal axis

Dufour & Meeks (2024) Ch 6

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Animations

Transition types



Figure 7.6 Easing function that mimic the elastic, overshoot, and bounce reactions. On each graph, the horizontal axis represents the time, while the vertical axis represents the change rate.

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Specifying transitions



Figure 7.7 D3 transitions affect only the properties chained after the `transition` method. We can customize transitions by setting their duration, delay, and easing function.

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Creating smooth transition

Transitions only work properties chained after the transition

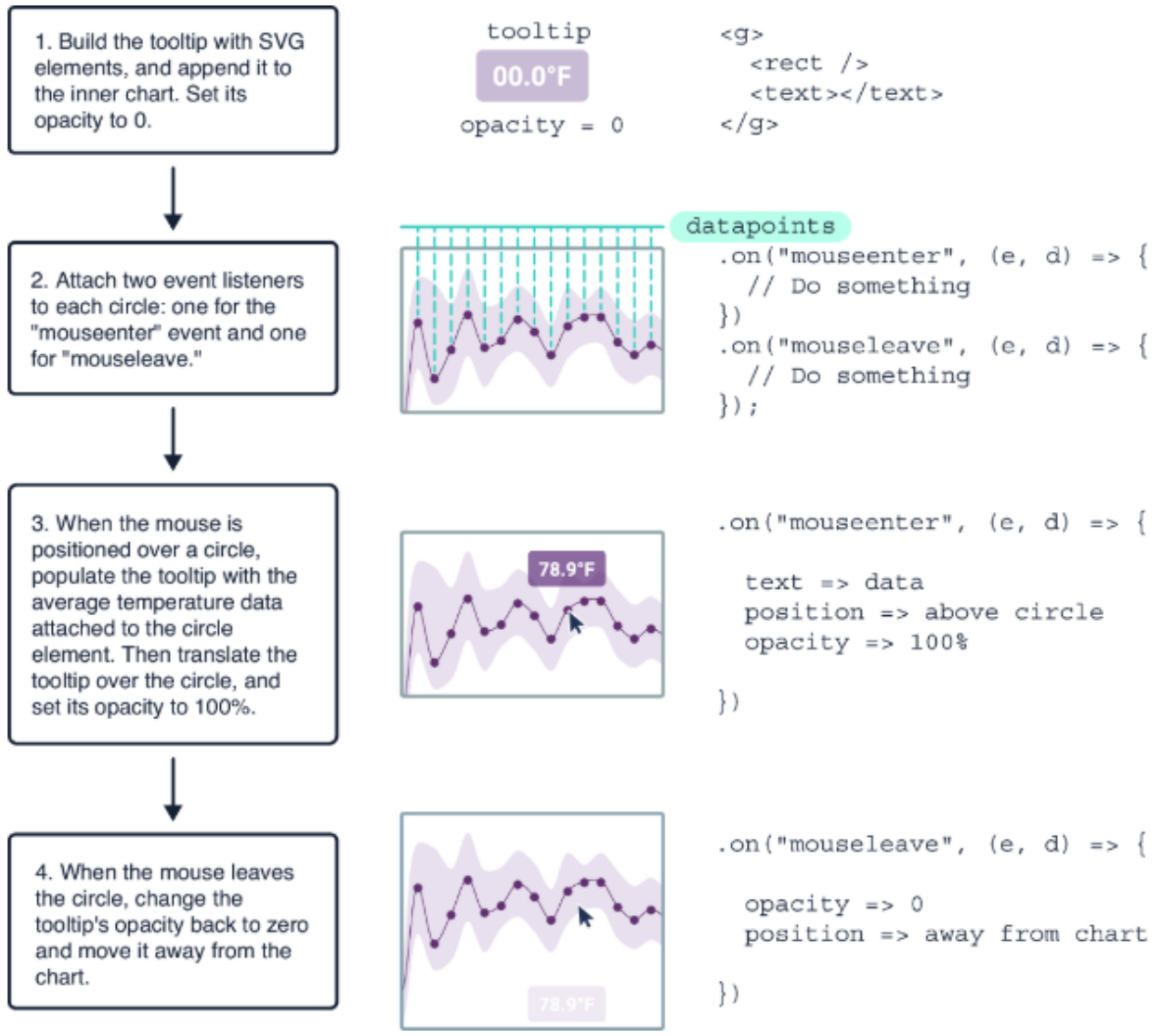
```
d3.selectAll("#histogram rect")
        .data(updatedBins)
        .transition()
                .duration(500)
                .ease(d3.easeCubicInOut)
                .attr("y", d => yScale(d.length))
                .attr("height", d => innerHeight - yScale(d.length));
```


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Tool Tips

Append group element

- Rectangle for background of tool tip
- Text for tool tip



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Set up

Add separate file for loading data

Include calls to our tooltip and handle mouse events

Make sure new files are all called in **index.html** (order is important)

```
<!-- Load your scripts here -->
<script src="https://d3js.org/d3.v7.min.js"></script>
<script src="js/shared-constants.js"></script>
<script src="js/interactions.js"></script>
<script src="js/line-chart.js"></script>
<script src="js/arcs.js"></script>
<script src="js/loading-data.js"></script>
```



loading-data.js > ...

```
d3.csv("../data/MelbTempData2024.csv", row => {
    return {
        date: d3.timeParse("%Y-%m-%d")(row.date),
        max_temp: +row.max_temp,
        min_temp: +row.min_temp,
        ave_temp: +row.ave_temp
    };
}).then(data => {
    console.log("temperature data", data);
    drawLineChart(data);
    createTooltip();
    handleMouseEvents();
});
```


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What to share in shared-constants.js

shared-constants.js

Move from line chart, gives our tool tips access to dimensions of chart

```
const margin = {top: 40, right: 170, bottom: 25, left: 40};
const width = 1000;
const height = 500;
const innerWidth = width - margin.left - margin.right;
const innerHeight = height - margin.top - margin.bottom;
const aubergine = '#6F2C91';

let innerChart;
```

```
const innerChart = svg
.append("g")
.attr("transform", `translate(${margin.left}, ${margin.top})`)
```

line-chart.js update

```
innerChart = svg
.append("g")
.attr("transform", `translate(${margin.left}, ${margin.top})`)
```

And delete aubergine const!

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Create tooltip

```
const createTooltip = () => {

const tooltip =innerChart
  .append("g")
    .attr("class", "tooltip")

}
```

Class name can be used later to select and manipulate tool tip

Tool tip will be visible at start, when completed testing set opacity to 0

```
    .style("opacity", 0);
```


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Add rectangles for tooltip and format

```
tooltip
    .append("rect")
    .attr("width", tooltipWidth)
    .attr("height", tooltipHeight)
    .attr("rx", 3)
    .attr("ry", 3)
    .attr("fill", aubergine)
    .attr("fill-opacity", 0.75);
```

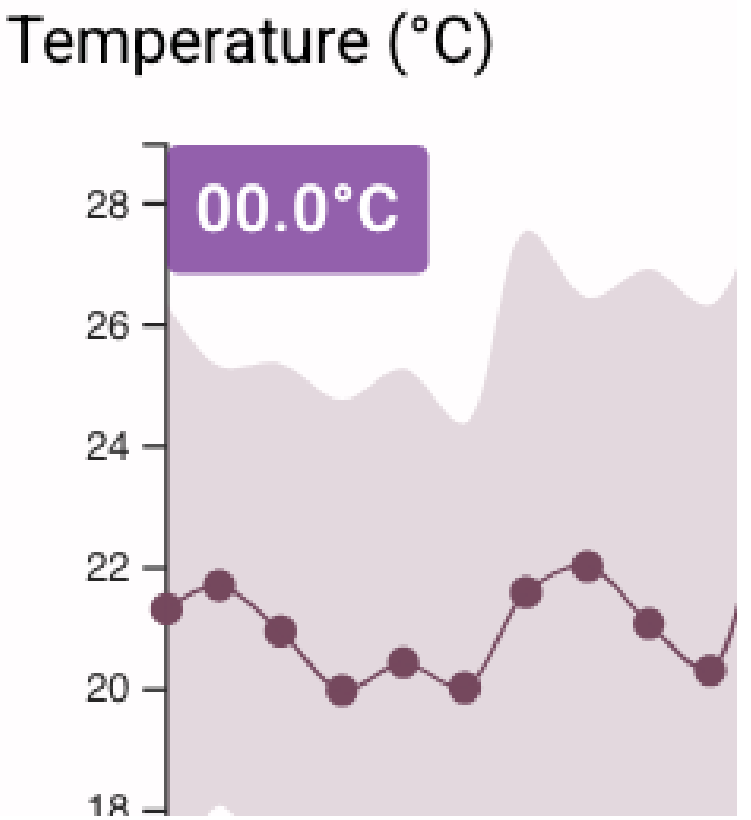
Add rounded edges

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Add text for tooltip and format

```
tooltip
    .append("text")
    .text("00.0°C")
    .attr("x", tooltipWidth/2)
    .attr("y", tooltipHeight/2 + 1)
    .attr("text-anchor", "middle")
    .attr("alignment-baseline", "middle")
    .attr("fill", "white")
    .style("font-weight", 900);
```

Placeholder test text



Add mouse interactivity

```
const handleMouseEvents = ()=> {

}
```

```
Const handleMouseEvents = ()=> {

        innerChart.selectAll("circle")
                .on("mouseenter", (e, d) => {
                        console.log("DOM event", e);
                        console.log("Attached datum", d);
                })
                .on("mouseleave", (e, d) => {

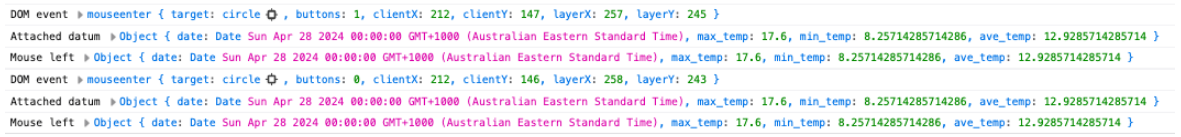
                });
}
```

Select all circles

Chain two D3 **.on** methods to detect when mouse entered and left circle

Load-data.js

```
d3.csv("../data/MelbTempData2024.csv", row => {
    return {
        date: d3.timeParse("%Y-%m-%d")(row.date),
        max_temp: +row.max_temp,
        min_temp: +row.min_temp,
        ave_temp: +row.ave_temp
    };
}).then(data => {
    console.log("temperature data", data);
    drawLineChart(data);
    createTooltip();
    handleMouseEvents();
});
```



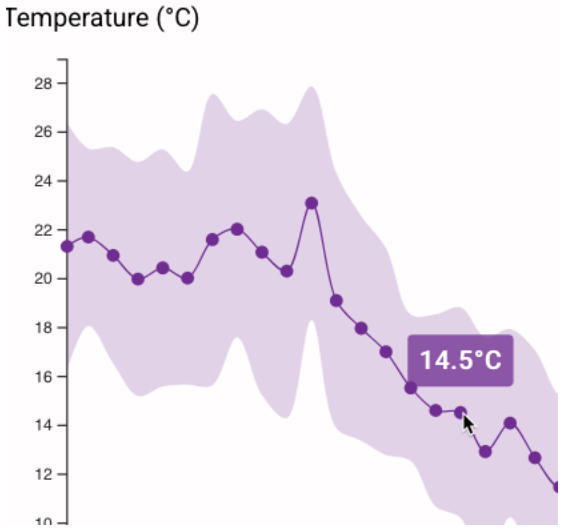
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Make tool tip appear

Set tool tip text to data found in circle element (eg temp)

Move tooltip above circle

Set tooltip opacity to 100%




```
innerChart.selectAll("circle")
    .on("mouseenter", (e, d) => {

        d3.select(".tooltip text")
            .text(`${d3.format(".3")(d.ave_temp)}°C`);

        const cx = e.target.getAttribute("cx");
        const cy = e.target.getAttribute("cy");

        d3.select(".tooltip")
            .attr("transform", `translate(${cx - 0.5*tooltipWidth}, ${cy -1.5*tooltipHeight})`)
            .transition()
            .duration(200)
            .style("opacity", 1);

    })
```

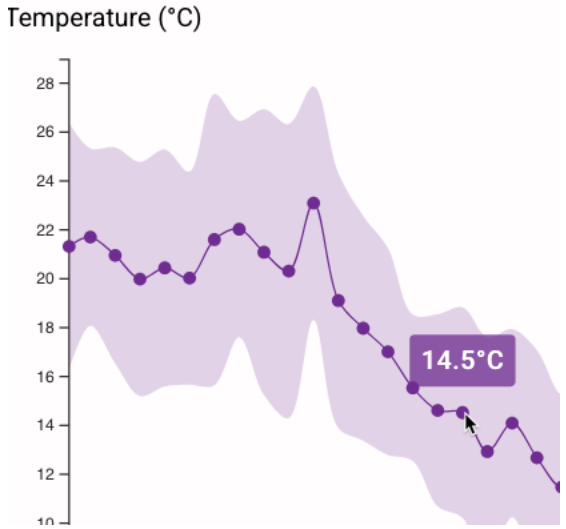

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Make tool tip disappear

Set tool tip text to data found in circle element (eg temp)

Move tooltip above circle

Set tooltip opacity to 100%




```
innerChart.selectAll("circle")
    .on("mouseenter", (e, d) => {

        d3.select(".tooltip text")
            .text(`${d3.format(".3")(d.ave_temp)}°C`);


        const cx = e.target.getAttribute("cx");
        const cy = e.target.getAttribute("cy");


        d3.select(".tooltip")
            .attr("transform", `translate(${cx - 0.5*tooltipWidth}, ${cy -1.5*tooltipHeight})`)
            .transition()
            .duration(200)
            .style("opacity", 1);
    })
    .on("mouseleave", (e, d) => {


        d3.select(".tooltip")
        .style("opacity", 0)
        .attr("transform", `translate(0, 500)`);
    });
```


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